Customer Segmentation & Sales Forecasting Analytics

An end-to-end Power BI project: data cleaning, customer groups and a 6-month sales forecast

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The business problem

A retail business has thousands of orders, but no clear answer to three simple questions.

1

Who are our most valuable customers?

Answered by: Customer segmentation

2

What will sales look like over the next 6 months?

Answered by: Sales forecasting

3

Which products and regions drive sales?

Answered by: Category, region and sub-category analysis

Solution: clean the data, group customers with DAX, forecast monthly sales, and show everything in one interactive Power BI report.

Dataset and tools

The data

- Superstore-style **sample retail dataset**, used to practise the full analytics workflow
- **1,015 raw records**, 15 columns, orders from Jan 2022 to Dec 2024
- Fields: Order ID and dates, customer, segment, region, city, category, sub-category, quantity, sales, discount, profit

The tools

Data cleaning

Remove duplicates and blanks, fix spelling, validate values

Power BI Desktop

Data model, segmentation, forecast, 3-page report

DAX

Customer summary table, segment rules, profit margin

GitHub

Hosts the report file, cleaned data and README

How the project was built

1

Raw data

1,015 records, 15 columns

2

Data cleaning

Fix errors, 971 clean rows

3

Segmentation

3 customer groups using DAX

4

Forecasting

Next 6 months, 95% confidence

5

Dashboard

3-page Power BI report on GitHub

Each step feeds the next: cleaned data becomes customer segments and a forecast, and both appear in the report.

Data cleaning

Issue found	Action taken	Rows
Duplicate records	Removed exact duplicates	**14**
Blank Sales values	Removed (Sales is a critical field)	**20**
Quantity of 0 or negative	Removed as data-entry errors	**10**
Inconsistent Region spelling	Standardized to North, South, East, West	**fixed**
Blank customer names	Labelled as "Unknown"	**fixed**
Dates stored as text	Converted to proper date type	**fixed**

Before

1,015

rows

44 rows removed

After

971

clean rows, 178 customers

Why it matters: wrong or missing values would give wrong totals, so cleaning comes first.

How customers were segmented

Segmentation groups customers who behave alike. The approach is **RFM-inspired**: Recency, Frequency, Monetary. This project uses **Frequency** (order count) and **Monetary** value (total sales).

High Value

Total sales above 3,000 and 5 or more orders

Regular

Total sales above 1,000

Low Value / At Risk

All remaining customers

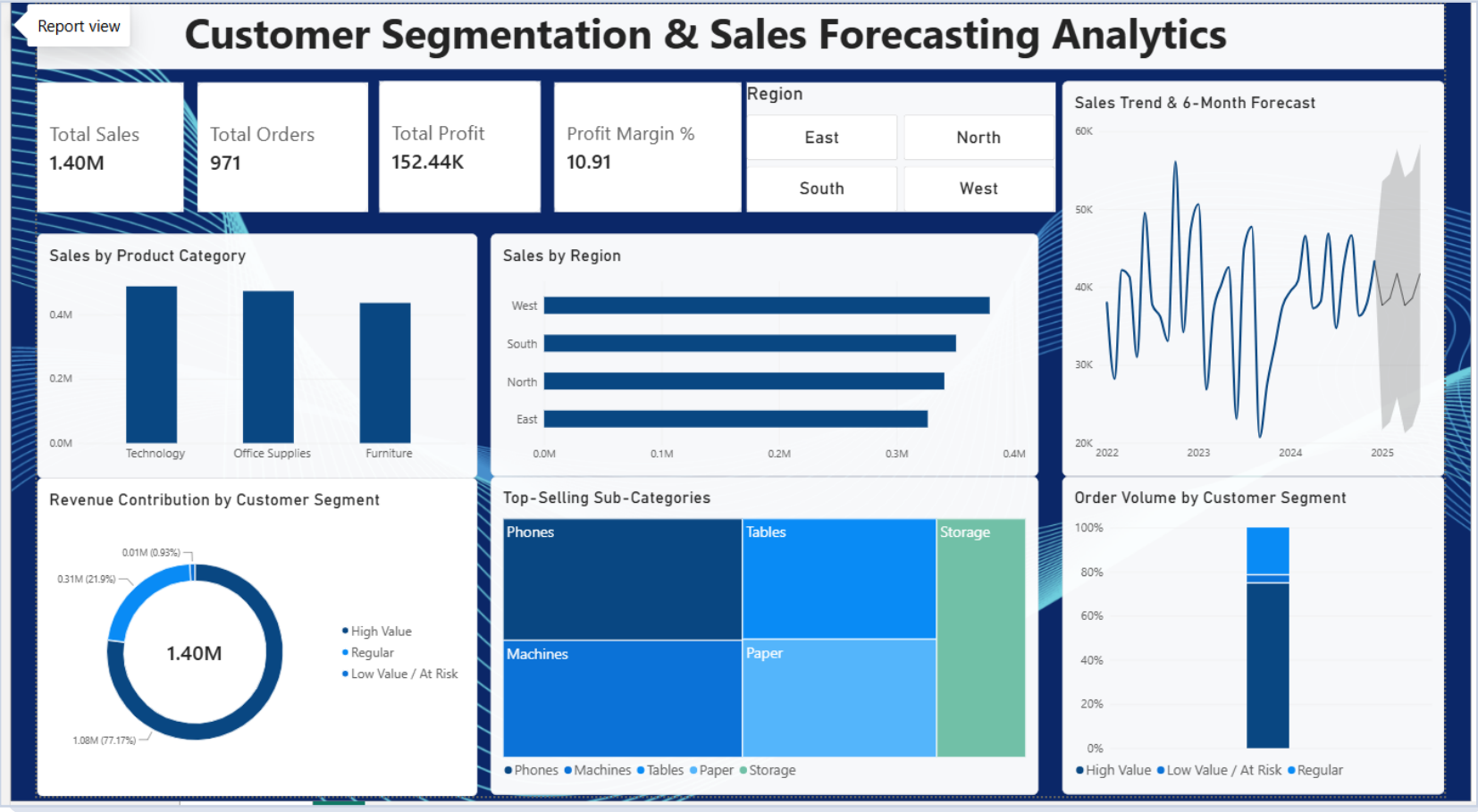
Step 1: one row per customer (DAX)

```
Customer_Summary = SUMMARIZE(
  superstore_sales,
  Customer_ID, Customer_Name,
  "Total_Sales", SUM(Sales),
  "Order_Count", COUNT(Order_ID),
  "Last_Order_Date", MAX(Order_Date))
```

Step 2: label each customer (calculated column)

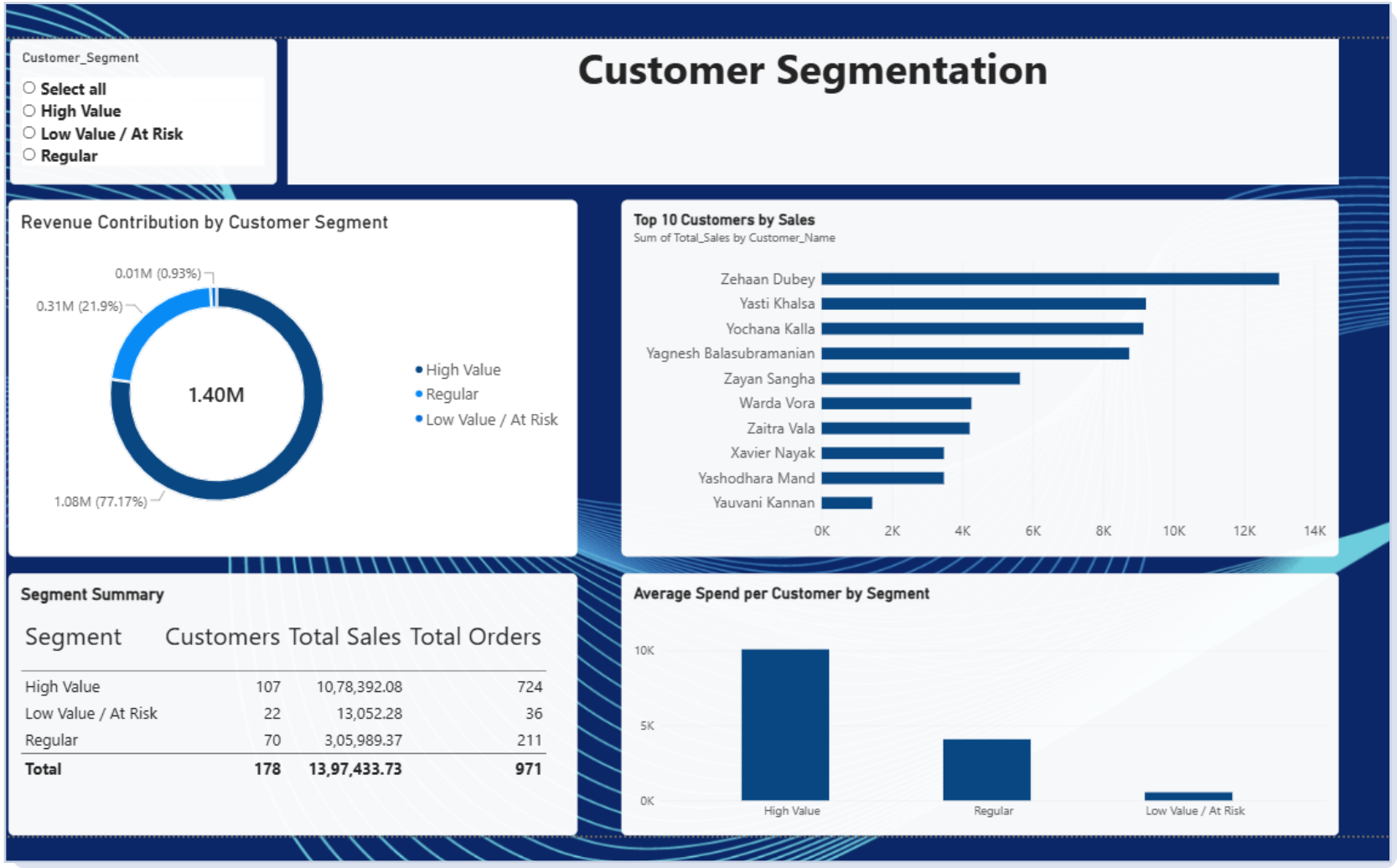
```
Customer_Segment =
IF(Total_Sales > 3000 && Order_Count >= 5,
  "High Value",
  IF(Total_Sales > 1000, "Regular",
  "Low Value / At Risk"))
```

Report page 1: Overview



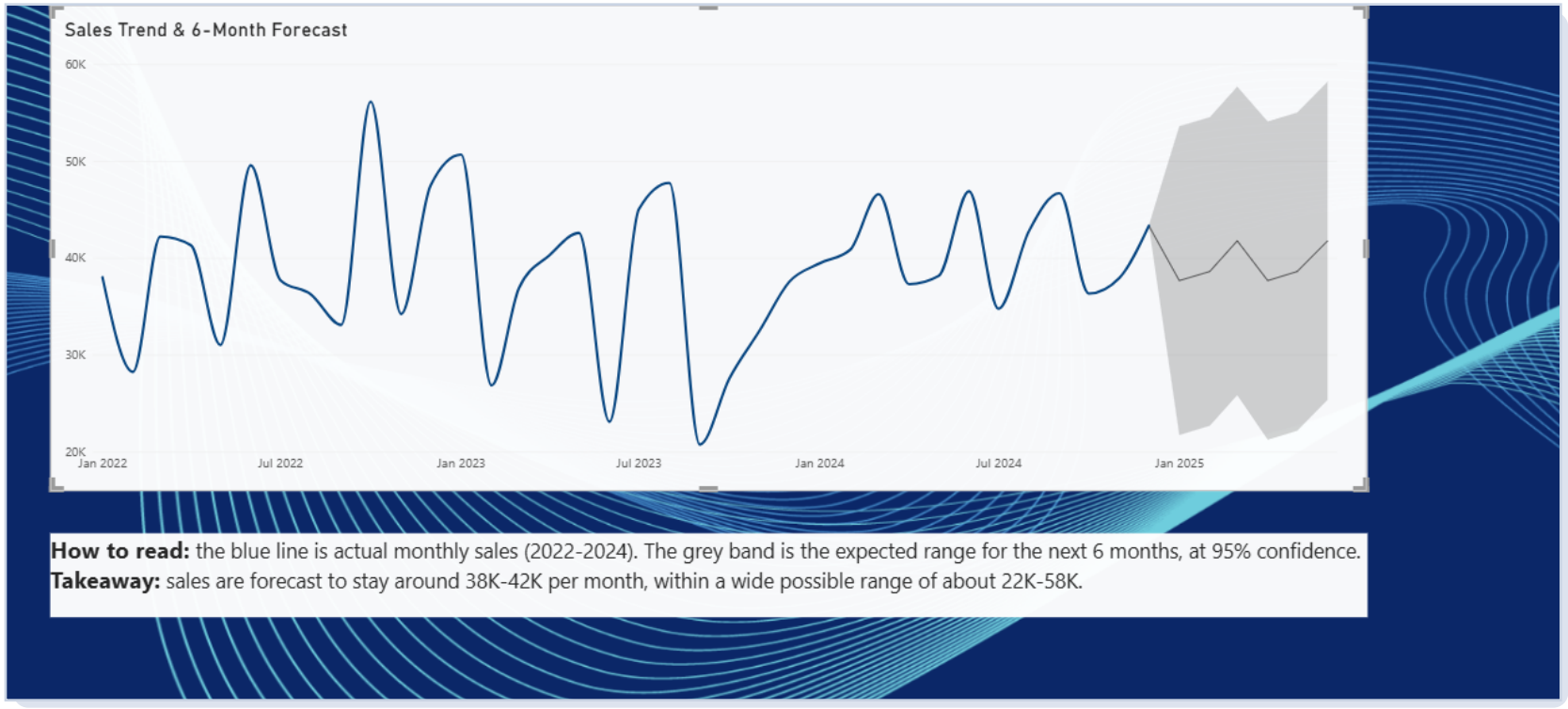
- **4 KPI cards:** Total Sales 1.40M, Total Orders 971, Total Profit 152.44K, Profit Margin 10.91%
- **Technology** is the top category and **West** is the top region
- **Phones** lead the sub-category sales
- The **Region slicer** filters the whole page

Report page 2: Customer Segmentation



- **107 of 178 customers (60%)** are High Value and bring **77.17%** of revenue
- Average spend per customer: High Value about **10.1K**, Regular about **4.4K**, Low Value / At Risk about **0.6K**
- The Top 10 list shows which customers to retain first

Report page 3: Sales Forecast



- **Solid line:** actual monthly sales, 2022 to 2024
- **Grey band:** expected range for the next 6 months, at 95% confidence
- Sales are forecast to stay around **38K to 42K** per month, within a wide range of about 22K to 58K
- Use: plan stock and promotions ahead

Key insights and recommendations

What the data shows

- **60% of customers** are High Value and generate **77.17%** of revenue
- A High Value customer spends about **2.3x** a Regular one and about **17x** a Low Value one
- **Technology** leads categories, **West** leads regions, **Phones** lead sub-categories
- Profit margin is **10.91%**, a baseline for pricing and discount decisions

What the business should do

- **Protect High Value customers** with loyalty rewards and priority service
- **Upgrade Regular customers** with targeted offers
- **Re-engage At Risk customers** with low-cost campaigns
- **Use the forecast** to plan stock and promotions

Limitations, future scope and conclusion

Limitations

- Sample dataset, so results demonstrate the method, not a real company
- Thresholds (3,000 and 5 orders) are simple rules; 60% of customers qualify as High Value, so a stricter rule would sharpen the focus
- Forecast uses 3 years of history only

Future scope

- Full RFM scoring that includes Recency
- Python machine learning for churn and forecasting
- Apply the same method to real company data

Conclusion: cleaned data, DAX segmentation and a 6-month forecast, in one interactive report. The main finding: High Value customers drive most of the revenue, so retention is the top priority.

Thank you

Questions?

Project files on GitHub

github.com/harish14Git/Customer-Segmentation-Sales-Forecasting

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